Lab 1: Fine-tuning a Transformer for Patent Phrase Matching

Ansh Upadhyay | Kaggle notebook: kaggle.com/code/anshupadhyay14/lab1-patent-matching (Version 2) | Model: microsoft/deberta-v3-small cross-encoder

Setup: anchor-grouped split (GroupShuffleSplit, SEED = 42, 20% of anchors) gives 28,914 train / 7,559 val rows (147 val anchors), used for every number below. Metric: Pearson *r*. Experiments ran on a local RTX 3050 GPU (fp16).

Task 1: Fine-tuned baseline

The starter notebook (3 epochs, lr 2e-5, batch 32, context = CPC section name) gives **val Pearson 0.8174** in 6.9 min. **Why hold out whole anchors?** Each anchor appears with about 50 targets. With a random-row split, the same anchors would be in train and val, so the model could memorise how each anchor is scored. The val score would be too high and would not predict the test score, because the test set has unseen anchors.

Task 2: Baseline without fine-tuning

Model	Fine-tuned?	Val Pearson
Bi-encoder all-MiniLM-L6-v2, cosine similarity	No	0.5814
Cross-encoder DeBERTa-v3-small (Task 1)	Yes	0.8174

A **bi-encoder** embeds anchor and target separately and compares two vectors, so the phrases never see each other or the context. A **cross-encoder** reads *[CLS] anchor [SEP] target; context [SEP]* as one input, so attention compares tokens directly, and a regression head learns the raters' scale. Fine-tuning added **+0.24** Pearson: patent vocabulary, use of context, and the 0 to 1 scale. Part of the gain is also from the architecture change.

Task 3: Controlled experiments (one change at a time)

Run	Change	Val r (seed 42)	Diff	Time	Interpretation
baseline	none	0.8174	0	6.9 min	reference
context=none	no context	0.8154	-0.0020	11.4 min	within noise
context=code	raw CPC code (G02)	0.8225	+0.0051	7.3 min	small, within seed noise
lr=1e-5	learning rate	0.8097	-0.0077	11.8 min	under-fits (train loss 0.035 vs 0.030)
lr=5e-5	learning rate	0.8252	+0.0078	7.1 min	best; checked with 2nd seed

Seed check	Seed 42	Seed 7	Mean	Mean diff vs baseline
baseline	0.8174	0.8227	0.8200	0
context=code	0.8225	0.8228	0.8226	+0.0026
lr=5e-5	0.8252	0.8244	0.8248	+0.0048

Conclusions: the baseline alone moves 0.0053 between seeds, so gaps under about 0.005 are noise. lr = 5e-5 beat both baseline seeds with both of its seeds, which makes it the most likely real (but small) gain. Context format (none / section name / raw code) shows **no reliable difference**. lr = 1e-5 is lower on one seed only. Training time varied 6 to 16 min for identical settings (laptop GPU throttling), so time differences are not caused by the settings.

Task 4: Error analysis (best model, lr = 5e-5)

4.1 Pearson per CPC section (val rows in brackets)

G Physics	C Chemistry	A Human nec.	D Textiles	H Electricity	E Constr.	F Mech. eng.	B Operations
0.709 (1040)	0.777 (1177)	0.791 (943)	0.805 (199)	0.809 (933)	0.810 (373)	0.852 (968)	0.900 (1926)

G (Physics) is hardest: a very broad section (optics, measuring, computing, instruments) with abstract terms. **C** has chemical names that a general model knows poorly. **B** has concrete mechanical phrases and the most data. With only 147 val anchors, per-section values (especially D) are noisy.

4.2 Mean prediction per true score

True score	0	0.25	0.5	0.75	1.0
Rows	1522	2465	2475	832	265
Mean pred	0.097	0.298	0.484	0.645	0.999

The model **compresses towards the middle** at 0 (pushed up to 0.10) and 0.75 (pulled down to 0.645). This matches the label imbalance: 0.25 and 0.5 are 65% of training labels, 0.75 only 11% and 1.0 only 3%, so MSE training favours middle values. Score 1.0 is not compressed, because those pairs are usually near-identical strings.

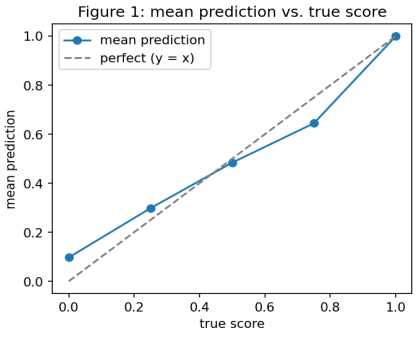


Figure 1: mean prediction vs. true score

4.3 Ten largest errors fall into two types:

(a) Shared words, true 0, model predicts high: punch face / face punch (pred 0.94), imaging axis / imaging of axis (0.85), boom hydraulic cylinder / boom a hydraulic cylinder (0.79), imaging axis / image of axis (0.76), return structure / return to structure (0.66). The model relies on word overlap; word order or a small word changes the meaning. Some labels are **debatable**: "boom a hydraulic cylinder" looks like a typo of the anchor.

(b) Few shared words, true 0.75, model predicts about 0: running tally / points balance, whose identity / customer id, breech mechanism / breech loading naval gun, end pins / wood pin, average pore size / average cell size. These need domain or field knowledge (pore = cell in foams; id = identity) that the small model lacks.

Task 5: Late submission and leaderboard scores

The final model **lr = 5e-5** was chosen by validation only, before any leaderboard score was seen. Kaggle notebook Version 1 crashed (*warmup_ratio* was removed in transformers 5) and produced no score. It was fixed with the equivalent *warmup_steps*, and **Version 2** is the only submission made.

#	Version	Model	Val r (Kaggle run)	Public	Private
1	Version 2	DeBERTa-v3-small, lr = 5e-5	0.8254	0.7966	0.8204

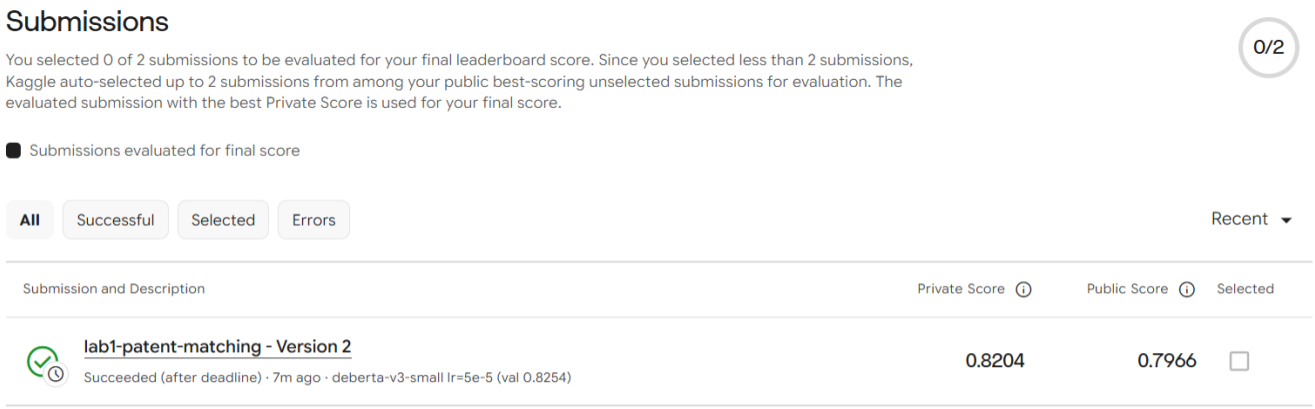


Figure 2: Kaggle Submissions tab (late submission)

Comparison: validation (0.825) and private (0.820) are very close, so the anchor-grouped split was a good estimate. Public is about 0.02 to 0.03 lower. **Reasons:** the public part of the test set is small, so its score is noisier; the test set has unseen anchors and a different section mix (section r ranges from 0.71 to 0.90); and choosing lr = 5e-5 by its validation score makes that score slightly optimistic.

References: Kaggle, U.S. Patent Phrase to Phrase Matching competition; Aslanyan & Wetherbee (2022), arXiv:2208.01171; He et al. (2021), DeBERTaV3, arXiv:2111.09543; Reimers & Gurevych (2019), Sentence-BERT, arXiv:1908.10084; Hugging Face Transformers and Sentence-Transformers docs; course starter notebook (pipeline structure and tokenizer fix adapted).